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(57) Abstract:

This invention provides a system and method for early detection of wildfires in remote forests. The system uses a network of self-sufficient solar-powered sensor nodes placed throughout the forest. Each node contains multiple sensors including a flame sensor, a smoke sensor, and a temperature and humidity sensor connected to a microcontroller. This microcontroller runs logic that cross-checks readings from all sensors before confirming a fire, which prevents false alarms. At the same time, every node has a long-range LoRa radio set up for peer-to-peer communication. When a fire is detected, the node broadcasts an alert. Neighbouring nodes automatically receive this alert and pass it along, creating a relay chain that carries the message through the forest without needing any towers or cables. This message eventually reaches a receiver station at the forest edge, which forwards it over the internet to a cloud server. The server then immediately notifies authorities through a mobile application and a web dashboard. The key innovation is the integration of accurate multi-sensor fire detection with a self-organizing wireless mesh network all packaged into a single low-cost solar-powered unit.

**FORM 2**  
**THE PATENTS ACT, 1970**  
**(39 OF 1970)**  
**&**  
**THE PATENTS RULES, 2003**  
**COMPLETE SPECIFICATION**  
**(See section 10 and rule 13)**  
**TITLE OF THE INVENTION**

**AUTONOMOUS WIRELESS MESH NETWORK SYSTEM FOR EARLY  
FOREST FIRE DETECTION IN REMOTE AREAS**

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## **PREAMBLE TO THE DESCRIPTION**

The following specification describes the invention and the manner in which it is to be performed.

## **FIELD OF THE INVENTION**

**[0001]** The present invention relates to the field of environmental monitoring and disaster prevention systems. More specifically, it relates to an early warning system for detecting forest fires, particularly in remote and inaccessible forest areas that lack cellular network coverage and electrical infrastructure. The system integrates a network of autonomous, solar-powered sensor nodes equipped with multiple environmental sensors, including flame, smoke, temperature, and humidity detectors.

**[0002]** It further employs Long Range (LoRa) wireless communication technology in a peer-to-peer mesh network configuration to enable reliable, low-power, long-distance data transmission. The system also incorporates Global Positioning System (GPS) modules for location tagging and cloud-based alerting through mobile applications and web dashboards. The primary application of this invention is in wildfire detection and prevention, but it may also be adapted for other environmental monitoring tasks in remote locations.

## **BACKGROUND OF THE INVENTION**

**[0003]** Forest fires are a serious problem worldwide. These fires destroy forests, damage property, and often cause loss of life. Weather patterns have changed over time and fires now occur more often and burn with greater intensity. Early detection

gives firefighters a chance to stop fires before they grow large. However, making this work in practice has proven difficult.

**[0004]** Different methods have been tried over the years. Satellites can scan large areas but they only pass over any given spot every few hours. Clouds also block their view. Aircraft and helicopters work well but flying them continuously costs too much money. Watchtowers with human observers have been used for many years. But people get tired, their vision has limits, and they cannot see well at night or in fog.

**[0005]** When the Internet of Things became popular, many thought it might offer a better way. Place many sensors in the forest that watch for problems day and night. These sensors would look for sudden heat, smoke, or gases that appear when a fire starts. But one big question remained: how does data from deep inside a forest reach someone who can respond?

**[0006]** Most wireless sensor networks today use short-range methods like Zigbee or Wi-Fi. To cover a large forest with these methods, one would need hundreds of nodes. These nodes would have to talk to each other in complex ways. This uses battery power quickly and makes installation very hard. Regular cell networks do not work because remote forests have no signal.

**[0007]** LoRa radio has features that look promising for this job. It can send small amounts of data over long distances while using little power. This makes it good for places with no electrical power. But most LoRa systems for fire detection use a star pattern where every sensor talks to one main gateway. If that gateway stops working, the whole network fails. Also, every sensor must be close enough to reach the gateway directly. Hills and thick trees often make this impossible.

**[0008]** So current systems have trade-offs. Some have good sensors but cannot send the data out. Others can communicate well but use simple sensors that give false alarms. What is needed is something that brings both together: good detection and reliable communication in one package that runs on solar power and does not cost too much.

## **OBJECTS OF THE INVENTION**

[0009] The main goal of this invention is to build a better early warning system for forest fires that avoids the problems found in current methods.

[0010] Another goal is to make this system work without needing any cell towers or internet lines deep in the forest. The system should be independent.

[0011] A further goal is to create a network that keeps working even if some parts fail. The nodes should find ways to talk to each other on their own and pass messages along so data can travel much farther than any single radio could reach.

[0012] Getting detection right matters a great deal. The system uses several types of sensors that check each other before sounding an alarm. This should reduce false alerts that make people ignore warnings.

[0013] These units will sit in remote forests for years at a time. A key part of the plan is to power them entirely with solar energy and manage that power carefully.

[0014] When a fire does get detected, the information must reach the right people quickly. Fire crews and coordinators should receive clear updates right away through simple apps and websites they can use on their phones or computers.

[0015] The system should use common parts that anyone can buy, keeping costs low and making repairs easier.

## **SUMMARY OF THE INVENTION**

[0016] In short, this invention provides a complete system and method for finding wildfires early. The core of the system is a set of self-sufficient sensor nodes placed across the forest. Each node has two main parts: a multi-sensor detection module and a mesh communication module. Both run on power from a small solar setup.

[0017] The detection module holds several (you should mention the details of the sensors) an infrared flame detector, a semiconductor sensor that detects smoke and gases, and a digital thermometer that also reads humidity. These connect to a small microcontroller. This chip runs logic that looks for agreement between sensors. It will not call fire just because the temperature rose on a hot day; it wants to see a flame signal together with either smoke or very high heat.

[0018] The communication side uses a LoRa radio also controlled by the same microcontroller. This radio talks with nearby nodes. The software on each node gives it two jobs: send out its own sensor readings, and listen for messages from nearby nodes and pass them along. When all nodes do this, they naturally create a mesh network where data hops from node to node.

[0019] Every piece of data, whether a routine status update or an urgent fire alert, moves through this chain until it reaches one or more receiver stations at the forest edge. These stations have their own LoRa receiver, a more powerful microcontroller, and a way to reach the internet through Wi-Fi or a 4G connection. Their job is to catch all packets coming out of the forest and send them up to a cloud server.

[0020] That cloud takes in raw data, sorts it into risk levels, saves everything in a database, and pushes live updates to the people who need them. This happens through a web-based map dashboard and a mobile app that can alert a firefighter's phone instantly.

[0021] Looking at the whole method, it involves putting nodes in place, letting them form their own mesh network, having them check sensors on a schedule, double-check any suspected fire, creating and sending an alert packet, bouncing that packet through the mesh to the gateway, sending it to the cloud, and finally notifying the response teams.

[0022] The real advance here is not just the sensors or the radio alone. It is how they work together smoothly in one solar-powered box. This combination finally makes reliable early warnings possible in the most isolated forests.

## **BRIEF DESCRIPTION OF DRAWINGS**

[0023] The invention will now be described with reference to the accompanying drawings.

[0024] FIG. 1 shows a block diagram of the overall system architecture for early wildfire detection. The figure illustrates a plurality of autonomous sensor nodes deployed across a forest area, peer-to-peer LoRa mesh communication links between the nodes, a LoRa gateway or receiver station located at the forest edge, a

cloud server, and user interfaces including a mobile application and a web dashboard.

**[0025]** FIG. 2 shows a detailed block diagram of a transmitter or sensor node. The figure illustrates the internal components of a single sensor node, including a solar panel, charge controller, rechargeable battery, first microcontroller, flame detection sensor, smoke detection sensor, temperature and humidity sensor, GPS module, and LoRa transceiver module.

**[0026]** FIG. 3 shows a block diagram of a receiver or base station. The figure illustrates the internal components of the receiver station, including a LoRa receiver, a power supply, a second microcontroller (ESP32), a buzzer, an LED indicator, and a serial port or display for diagnostics.

**[0027]** FIG. 4 shows a flowchart of the operational logic of a sensor node. The figure illustrates the decision-making process including system initialization, low-power sleep mode, wake-up cause determination (timer interrupt or radio interrupt), sensor data acquisition (flame, smoke, temperature, humidity, GPS), fire condition evaluation using multi-sensor validation logic, data packet formation, LoRa transmission, mesh networking state for packet forwarding, and continuous cycle return to sleep mode.

## **DETAILED DESCRIPTION OF THE INVENTION**

**[0028]** A detailed description of how this system works now follows, using the drawings as reference. This describes one preferred embodiment. The core ideas could be implemented in other ways and still capture the essence of the invention.

### **Overall Architecture (FIG. 1)**

**[0029]** Consider a large remote forest that needs monitoring. Many identical self-powered sensor nodes go out across the area. The drawing shows three as examples: Node A (101), Node B (102), and Node C (103). In a real deployment you would use as many as needed to cover the ground. The important part is how they talk to each other. The arrows marked peer-to-peer mesh communication (104) show they do not all report to one central tower. Instead, they talk among themselves. Node A can reach Node B, which can reach Node C, making a web of connections.

**[0030]** This setup means data does not get trapped. If Node B (102) detects a fire, its alert does not need a direct line to the base station. It can pass the message to Node A (101), which then passes it to the LoRa Gateway or Receiver Station (105). This hopping does two useful things: it stretches the network reach far beyond what any single radio could manage, and it builds in backup paths. If one node fails or something blocks its path, the data can find another route through the web.

**[0031]** The LoRa Gateway or Receiver Station (105) sits somewhere with internet access, such as a ranger outpost at the forest edge. It acts as the bridge between the LoRa mesh (104) out in the woods and the online world. It receives every packet that makes its way out of the forest.

**[0032]** Once collected, the station (105) sends everything to the User Alert System (106) which runs on cloud servers. This system serves different users in different ways. The Mobile Alert (107) is an app on a firefighter's phone. The Web Dashboard (108) is a website for command center staff. The Emergency Alert (109) is the ability to trigger wider warnings like community SMS alerts or sirens.

### **Sensor Node (FIG. 2)**

**[0033]** Now look inside a sensor node. FIG. 2 shows the inside of one unit (200). It is built to be tough and weatherproof, usually mounted on a tree trunk. Since there is no power outlet in the forest, it makes its own. A small solar panel (201) of 6V and 10W captures sunlight. An MPPT solar charge controller (202) pulls the most power from the panel and safely charges a lithium-ion battery (203) (an 18650 cell). This battery keeps things running through the night. Other circuits clean up this power to steady 3.3V and 5V for the electronics.

**[0034]** The node is controlled by a first microcontroller (204). An Arduino Nano works well. It is small, has enough processing power, and can read both analog and digital signals. It decides when to check sensors, what the readings mean, when to use the radio, and when to relay a neighbour's message.

**[0035]** This microcontroller (204) connects to a set of specialized sensors. The flame sensor (205) is an LM393-based module with an infrared detector tuned to flame flicker. It gives a simple digital yes/no signal. The smoke sensor (206) is an MQ-2 semiconductor sensor whose resistance changes in the presence of smoke or



gases, producing an analog voltage. The temperature and humidity sensor (207) is a DHT22, sending precise digital readings over a single wire.

**[0036]** A NEO-8M GPS module (208) gives the node its location. It communicates with the microcontroller over a serial connection, providing latitude and longitude. Every reading is tagged with a location, which is vital for finding a fire.

**[0037]** For long-range communication, a LoRa transceiver module (209) based on the SX1262 chip connects to the microcontroller via SPI. It operates in the license-free ISM band (e.g., 866 MHz in India) with Spreading Factor set to 10 and Bandwidth set to 125 kHz, chosen to maximize range and tree penetration.

### **Receiver Station (FIG. 3)**

**[0038]** FIG. 3 shows the receiver station (300). It does not carry sensors. Its only job is to listen and forward. It has a matching LoRa receiver (301), a steady power supply (302) (often grid-connected), and a second microcontroller (303). An ESP32 is a good choice because it has Wi-Fi built in. The LoRa receiver (301) connects to the ESP32 via SPI.

**[0039]** The ESP32 (303) runs a straightforward program: keep listening on the LoRa frequency, grab any incoming packet, wrap the data into a JSON package, and use its Wi-Fi to POST it to a specific cloud server endpoint. For local awareness, the station may have a buzzer (305) and an LED (306) that sound and flash when a fire packet arrives. A small screen or serial port (307) helps with setup and troubleshooting.

### **Operational Flowchart (FIG. 4)**

**[0040]** The real intelligence sits in how the sensor node operates. FIG. 4 shows this as a flowchart, essentially the code running on the first microcontroller (204). It starts up and initializes all its parts - pins, sensors, radio - then enters its main power-saving routine.

**[0041]** It spends most of its time in a Low-Power Sleep Mode, using very little energy. It wakes up either when its own internal timer goes off (every 5 minutes) or when the LoRa radio signals that a packet has arrived.

**[0042]** If the timer wakes it, it acquires sensor data: it reads the flame detector, samples the smoke sensor analog voltage, reads temperature and humidity from the

DHT22, and fetches GPS coordinates. This whole snapshot takes about two seconds.

**[0043]** Now comes the fire decision. The logic is: Flame Sensor equals DETECTED AND (Smoke Reading greater than THRESHOLD OR Temperature greater than THRESHOLD). In other words, a flame must be present, and either smoke or heat must be elevated. Based on controlled test fires, the thresholds are set as: Smoke analog reading greater than 1200 (on a 10-bit ADC scale) or Temperature greater than 45 degrees Celsius.

**[0044]** If the condition is false, the node packages the data as a Normal Status. If true, it packages as a FIRE ALERT Status. The package includes node ID, status flag, all sensor readings, and GPS coordinates. Next, the node transmits the data packet via LoRa. For a FIRE ALERT, this injects the alert into the mesh network.

**[0045]** If the node woke up because of a received packet, it enters the LoRa Mesh Networking State. It listens, checks the packet's unique ID against a local cache, and if the packet is new and signal strength is adequate, it forwards the packet and updates its cache. This "hear-check-forward" rule, running on every node, builds the self-forming mesh.

**[0046]** After either transmission or forwarding, the flow returns to main listening and relay logic and then back to sleep, completing the continuous cycle. This design keeps power consumption low while enabling immediate relay of urgent alerts.

### **Working Example**

**[0047]** In a preferred embodiment, 100 sensor nodes are deployed in a 1 km by 1 km forest grid, spaced approximately 200 to 300 meters apart. Each node's microcontroller is programmed with a duty cycle of 5 minutes. During a controlled test fire, Node 47 located near the fire source detected a flame and a smoke analog reading of 1850.

**[0048]** The microcontroller set the FIRE ALERT flag and transmitted a packet. Node 32, located 150 meters away, received the packet, verified it was new, and retransmitted it. Within 15 seconds, the packet hopped through four intermediate nodes and reached the receiver station at the forest edge. The receiver forwarded the JSON data to the cloud server, which triggered an SMS and a mobile push

notification to the fire station. The entire end-to-end delay from fire ignition to alert was under 30 seconds.

### **Best Method**

[0049] The inventors consider the embodiment described above, with the specific component choices including Arduino Nano, LM393 flame sensor, MQ-2 smoke sensor, DHT22 temperature and humidity sensor, NEO-8M GPS, SX1262 LoRa module, and ESP32 gateway, and the threshold values including 5-minute sleep, 45 degrees Celsius temperature threshold, and smoke ADC threshold of 1200, to be the best method currently known for practicing the invention.

### **Cloud Server and Alerting**

[0050] The cloud server receives data from the receiver station via HTTPS POST. A backend program written in Python or PHP validates the request, unpacks the JSON, and applies classification rules. Normal status means temperature up to 35 degrees Celsius and humidity at least 50 percent and smoke ADC reading up to 1190. Surveillance status means temperature between 35 and 45 degrees Celsius or smoke ADC reading between 1190 and 1984. Fire status means temperature above 45 degrees Celsius or smoke ADC reading above 1984.

[0051] All data are stored in a MySQL database. The Web Dashboard subscribes to a live stream and plots node locations on a map using Leaflet or Google Maps. Marker colours indicate status: green for Normal, orange for Surveillance, red for Fire. The Mobile App receives push notifications via Firebase Cloud Messaging whenever a node's status changes to Surveillance or Fire. The notification includes node ID, GPS coordinates, and current sensor readings.

[0052] It should be understood that the embodiments described above are just one way to build this invention. A person skilled in the art could make modifications and variations without leaving the scope of the invention. The invention itself is defined by the claims that follow.

## CLAIMS

1. An early wildfire detection system comprising: a plurality of autonomous sensor nodes configured for deployment in a forest area, each sensor node including a flame detection sensor , a smoke detection sensor , a temperature and humidity sensor , a first microcontroller connected to said sensors, a LoRa transceiver module configured for peer-to-peer communication, and a solar power system having a solar panel , a rechargeable battery , and a power management circuit a receiver station located remote from said sensor nodes and including a LoRa receiver a second microcontroller and an internet connectivity module; wherein the plurality of sensor nodes are configured to autonomously form a peer-to-peer wireless mesh network using their respective LoRa transceiver modules such that a data packet generated by any sensor node is relayed by one or more intermediate sensor nodes to reach the receiver station .
2. The system of claim 1, wherein the first microcontroller is configured to execute a detection algorithm that requires a positive indication from the flame detection sensor concurrently with an elevated reading from at least one of the smoke detection sensor or the temperature sensor to trigger a fire alert status.
3. The system of claim 1, wherein each sensor node operates in a duty-cycled manner alternating between a low-power sleep mode and an active mode for sensor data acquisition and communication.
4. The system of claim 1, wherein the first microcontroller is configured to: (a) upon detection of a fire condition, generate and transmit an alert data packet using its LoRa transceiver and (b) independently and concurrently, listen for data packets from other sensor nodes and, upon receiving a new packet, automatically retransmit said packet to extend network range.
5. The system of claim 1, further comprising a cloud server in communication with the receiver station via the internet, said cloud server configured to: (a) receive and store data from the sensor nodes; (b) classify fire risk based on predefined thresholds applied to the received sensor data; and (c) generate and disseminate alerts to one or more user devices.

6. The system of claim 1, wherein the flame detection sensor is an LM393-based infrared flame sensor (205), and the smoke detection sensor is a semiconductor gas sensor, and the temperature and humidity sensor is a DHT22 digital sensor.
7. The system of claim 1, wherein each sensor node further comprises a GPS module configured to provide location coordinates, and wherein every transmitted data packet includes said location coordinates.
8. A method for early detection of wildfires using the system of any preceding claim, comprising the steps of: deploying a plurality of sensor nodes in a forest area; autonomously forming a peer-to-peer LoRa mesh network among said sensor nodes; periodically acquiring at each sensor node environmental data from a plurality of sensors processing said acquired data at the sensor node using a multi-sensor validation logic to detect a potential fire condition; if a fire condition is detected, generating an alert data packet at the detecting node and transmitting it via its LoRa transceiver relaying said alert data packet through one or more intermediate sensor nodes in the mesh network until it is received by a receiver station forwarding, by the receiver station, the alert data to a cloud server via the internet; and generating and disseminating, by the cloud server, a fire alert to one or more user interfaces
9. The method of claim 8, wherein the step of relaying comprises each intermediate sensor node listening for packets, verifying the packet is new, and automatically retransmitting it without requiring instruction from a central controller.
10. The method of claim 8, wherein the multi-sensor validation logic requires a concurrent positive reading from a flame sensor and an elevated reading from at least one of a smoke sensor or a temperature sensor.

## **ABSTRACT**

### **AUTONOMOUS WIRELESS MESH NETWORK SYSTEM FOR EARLY FOREST FIRE DETECTION IN REMOTE AREAS**

This invention provides a system and method for early detection of wildfires in remote forests. The system uses a network of self-sufficient solar-powered sensor nodes placed throughout the forest. Each node contains multiple sensors including a flame sensor, a smoke sensor, and a temperature and humidity sensor connected to a microcontroller. This microcontroller runs logic that cross-checks readings from all sensors before confirming a fire, which prevents false alarms. At the same time, every node has a long-range LoRa radio set up for peer-to-peer communication. When a fire is detected, the node broadcasts an alert. Neighbouring nodes automatically receive this alert and pass it along, creating a relay chain that carries the message through the forest without needing any towers or cables. This message eventually reaches a receiver station at the forest edge, which forwards it over the internet to a cloud server. The server then immediately notifies authorities through a mobile application and a web dashboard. The key innovation is the integration of accurate multi-sensor fire detection with a self-organizing wireless mesh network all packaged into a single low-cost solar-powered unit.